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OpenManus详解

[OpenManus](#) 是 MetaGPT 团队耗时仅 **3小时** 开发的开源 AI Agent 框架旨在复刻商业产品 Manus 的核心功能。

维度	OpenManus	Manus AI
运行环境	本地 Terminal 交互	云端虚拟电脑全流程可视化
工具复杂度	基础工具链（Python/浏览器/文件）	自研多模态工具组合
模型架构	单智能体 + 通用大模型（如 GPT-4o）	多智能体系统 + 精调专用模型
任务容错率	需手动调整 System Prompt ⁺ 优化	内置动态学习与错误自修复机制
部署成本	免费开源	需购买订阅服务（基础版 \$99/月）

生产 X
demo

思想 架构

123 → [str] → LLM → [str] json → %c name parameter

Instruction: Determine the most cost-effective country to purchase the smartphone model "CodeAct 1". The countries to consider are the USA, Japan, Germany, and India.

Available APIs

- [1] lookup_rates(country: str) -> (float, float)
- [2] convert_and_tax(price: float, exchange_rate: float, tax_rate: float) -> float

- [3] estimate_final_price(converted_price: float, shipping_cost: float) -> float
- [4] lookup_phone_price(model: str, country: str) -> float
- [5] estimate_shipping_cost(destination_country: str) -> float

LLM Agent using [Text/JSON] as Action

Think I should calculate the phone price in USD for each country, then find the most cost-effective country.

Action Text: lookup_rates, Germany
JSON: {"tool": "lookup_rates", "country": "Germany"}

Environment 1.1, 0.19

Action Text: lookup_phone_price, CodeAct 1, Germany
JSON: {"tool": "lookup_phone_price", "model": "CodeAct 1", "country": "Germany"}

Environment 700

Action Text: convert_and_tax, 700, 1.1, 0.19
JSON: {"tool": "convert_and_tax", "price": 700, "exchange_rate": 1.1, "tax_rate": 0.19}

Environment 916.3

[... interactions omitted (look up shipping cost and calculate final price) ...]

Action Text: lookup_rates, Japan
JSON: {"tool": "lookup_rates", "country": "Japan"}

[... interactions omitted (calculate final price for all other countries) ...]

Response The most cost-effective country to purchase the smartphone model is Japan with price 904.00 in USD

Fewer Actions Required!

CodeAct: LLM Agent using [Code] as Action

Think I should calculate the phone price in USD for each country, then find the most cost-effective country.

Action

```
countries = ['USA', 'Japan', 'Germany', 'India']
final_prices = {}

for country in countries:
    exchange_rate, tax_rate = lookup_rates(country)
    local_price = lookup_phone_price("xAct 1", country)
    converted_price = convert_and_tax(
        local_price, exchange_rate, tax_rate
    )
    shipping_cost = estimate_shipping_cost(country)
    final_price = estimate_final_price(converted_price, shipping_cost)
    final_prices[country] = final_price

most_cost_effective_country = min(final_prices, key=final_prices.get)
most_cost_effective_price = final_prices[most_cost_effective_country]
print(most_cost_effective_country, most_cost_effective_price)
```

Environment 1.1, 0.19

Response The most cost-effective country to purchase the smartphone model is Japan with price 904.00 in USD.

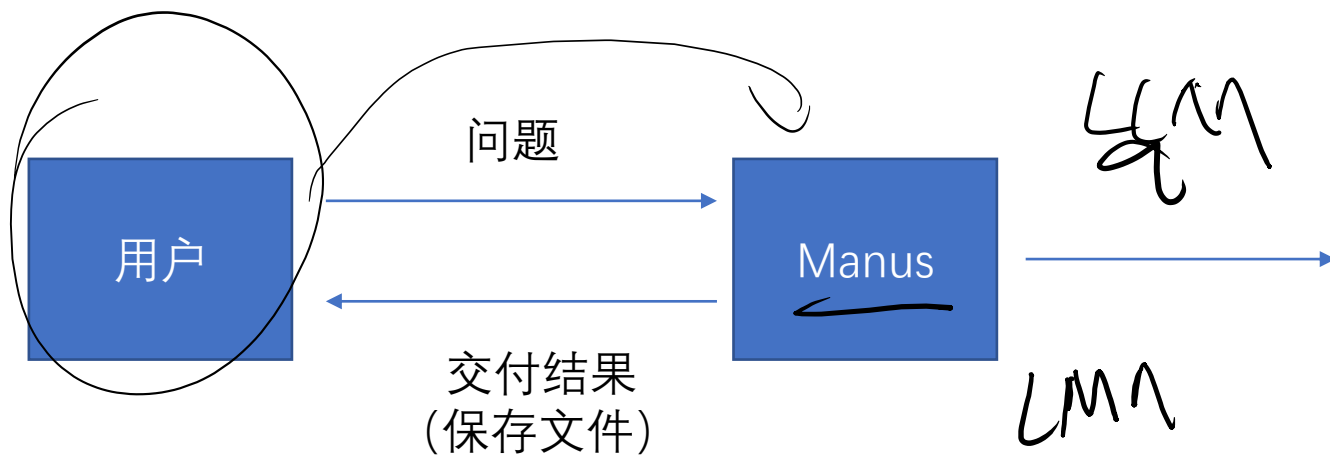
Control & Data Flow of Code
Simplifies Complex Operations

Re-use `min` Function from Existing
Software Infrastructures (Python library)

知乎 @东波先生

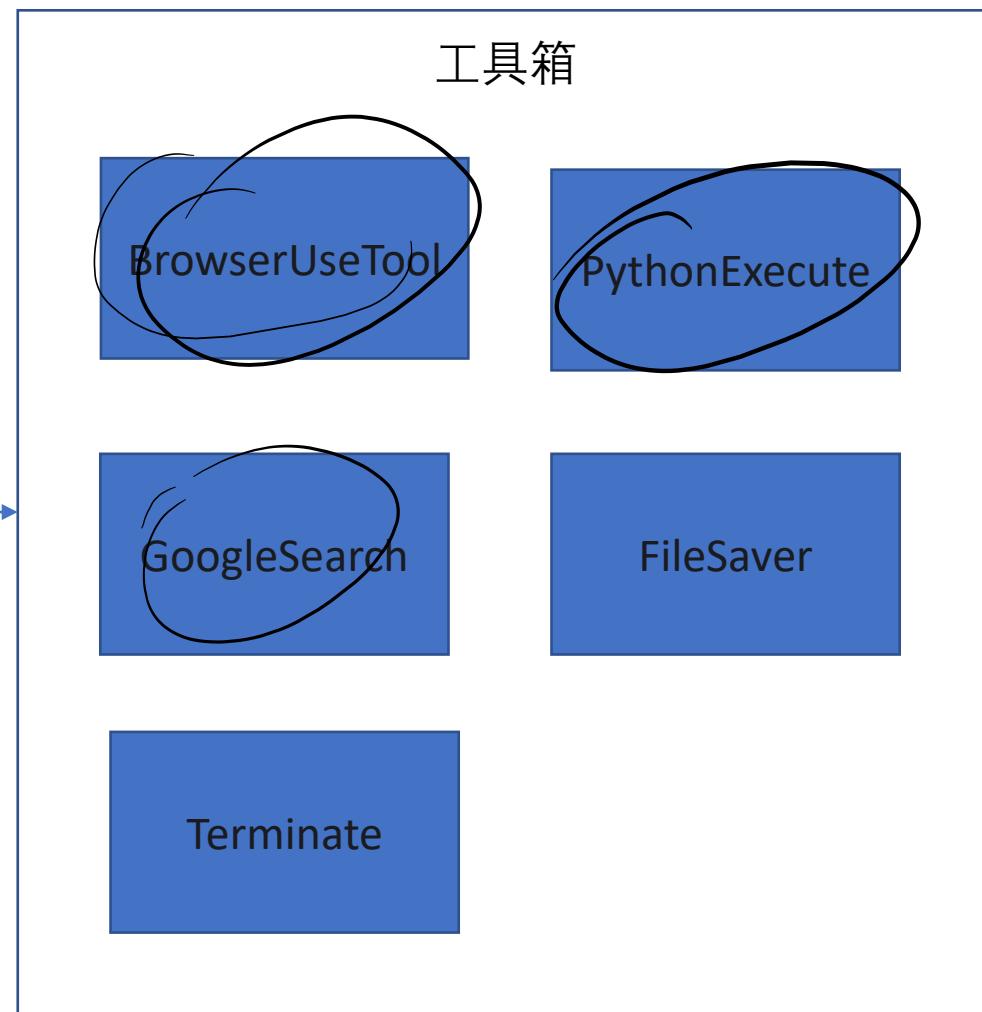
step → step ... → 结束 → 输出

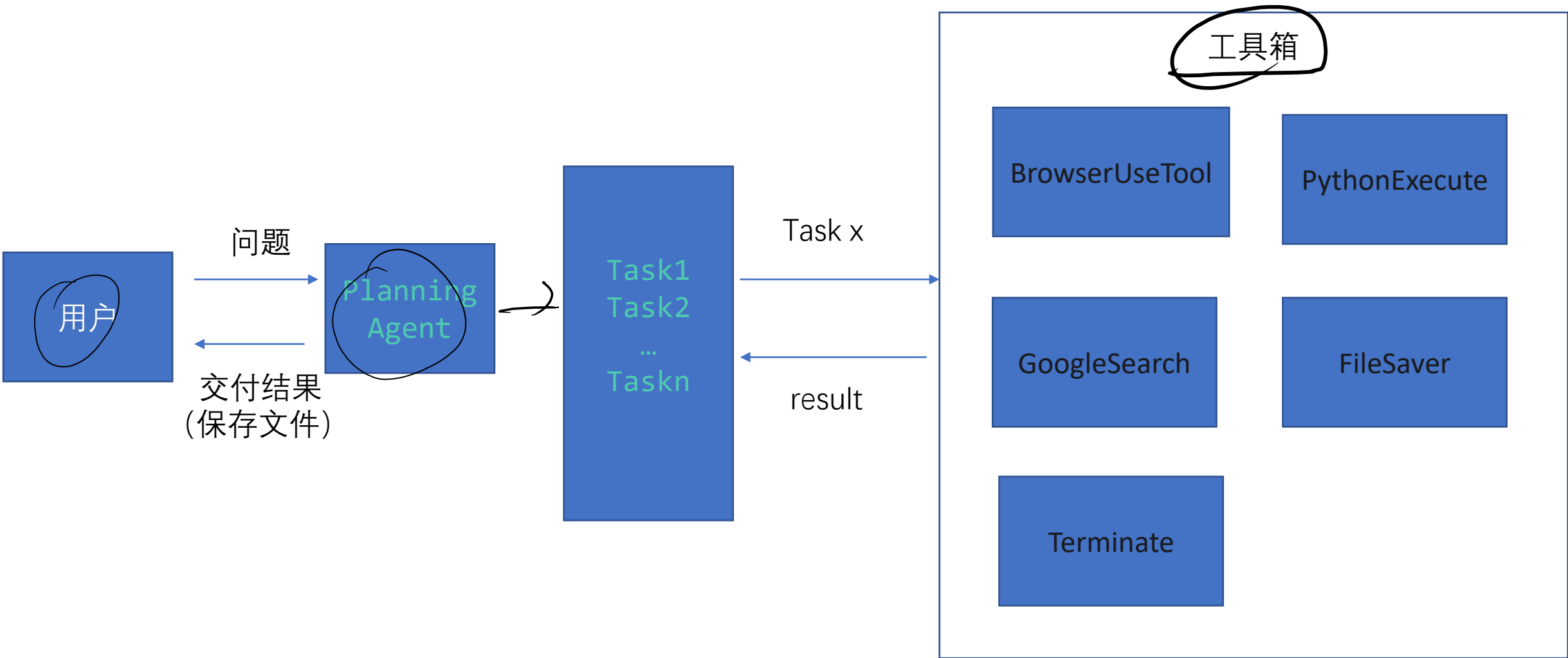
单智能体



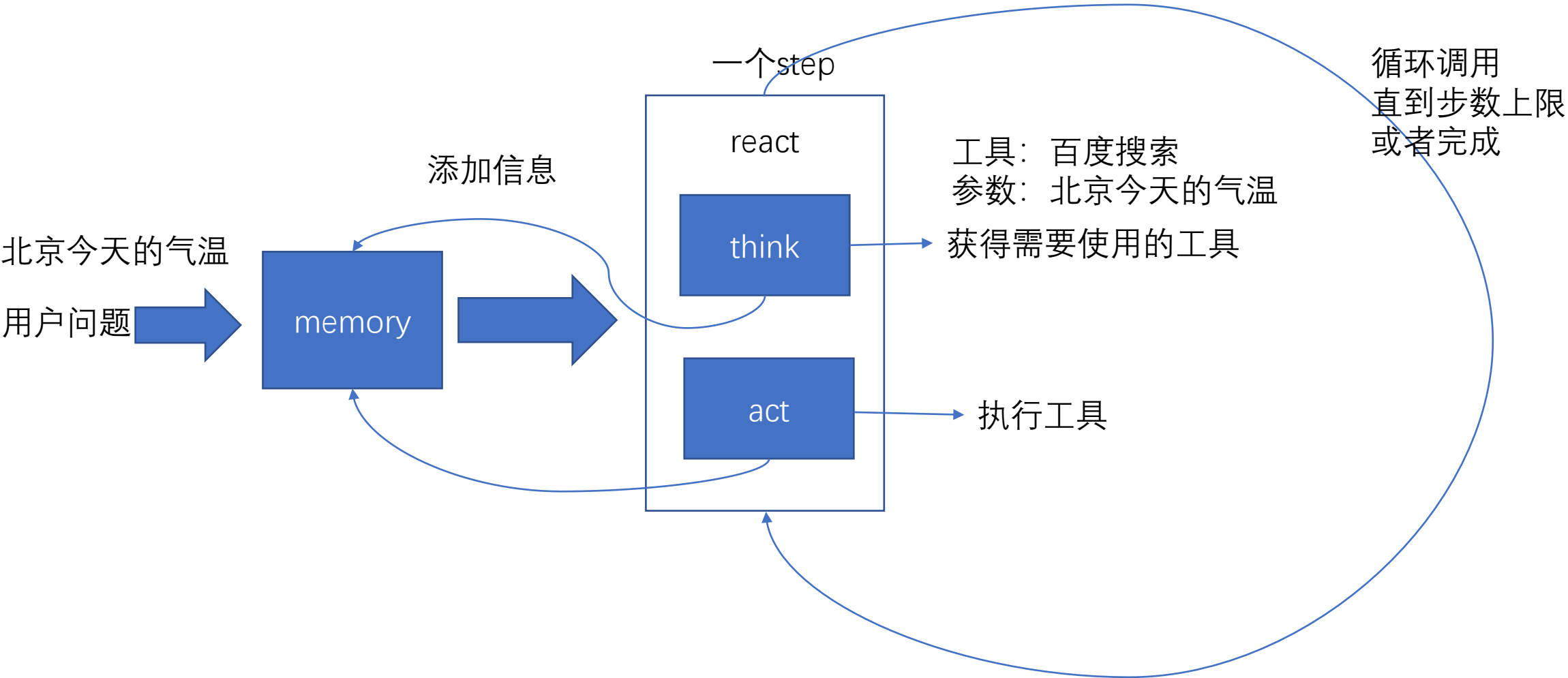
LLM

LLM





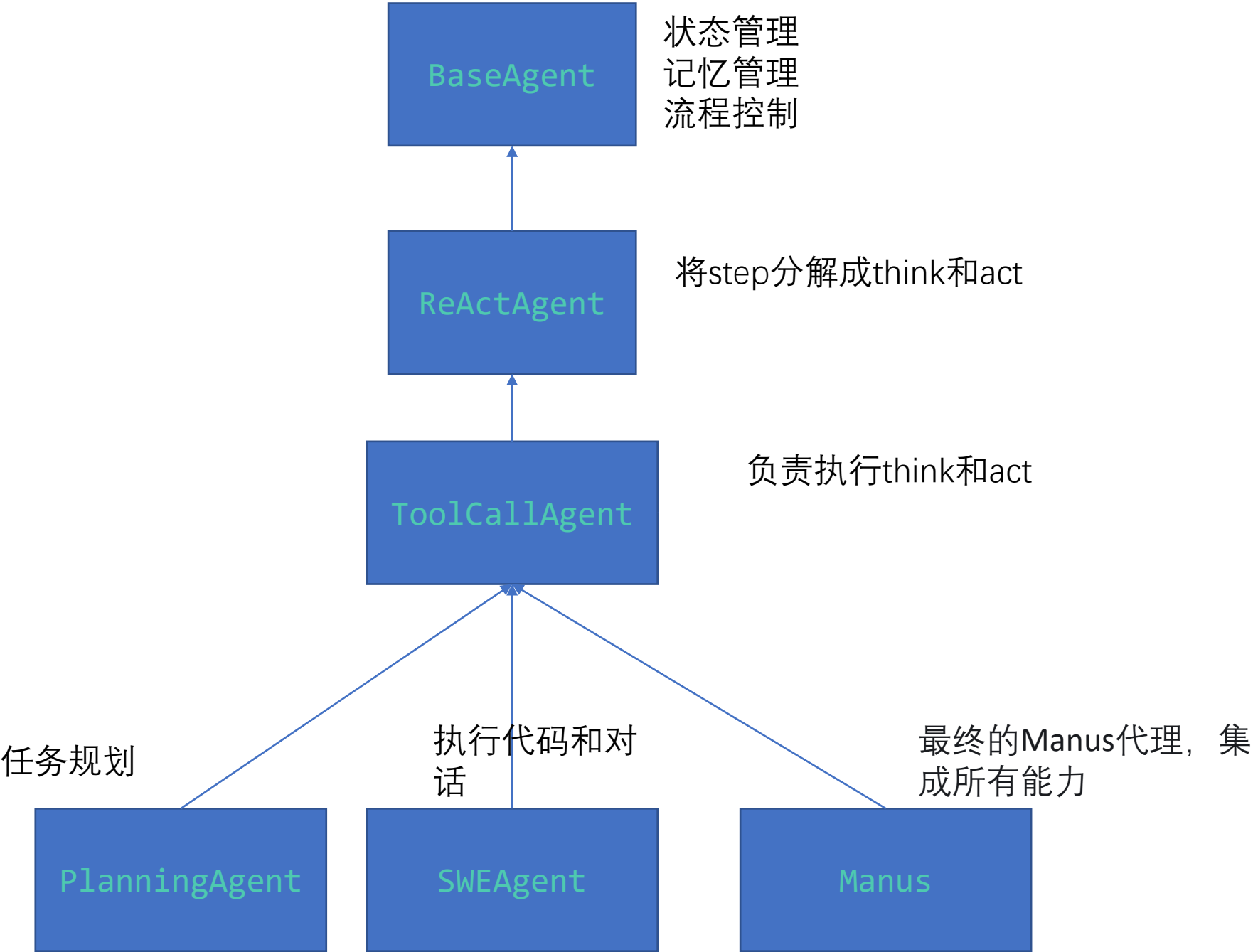
运行流程



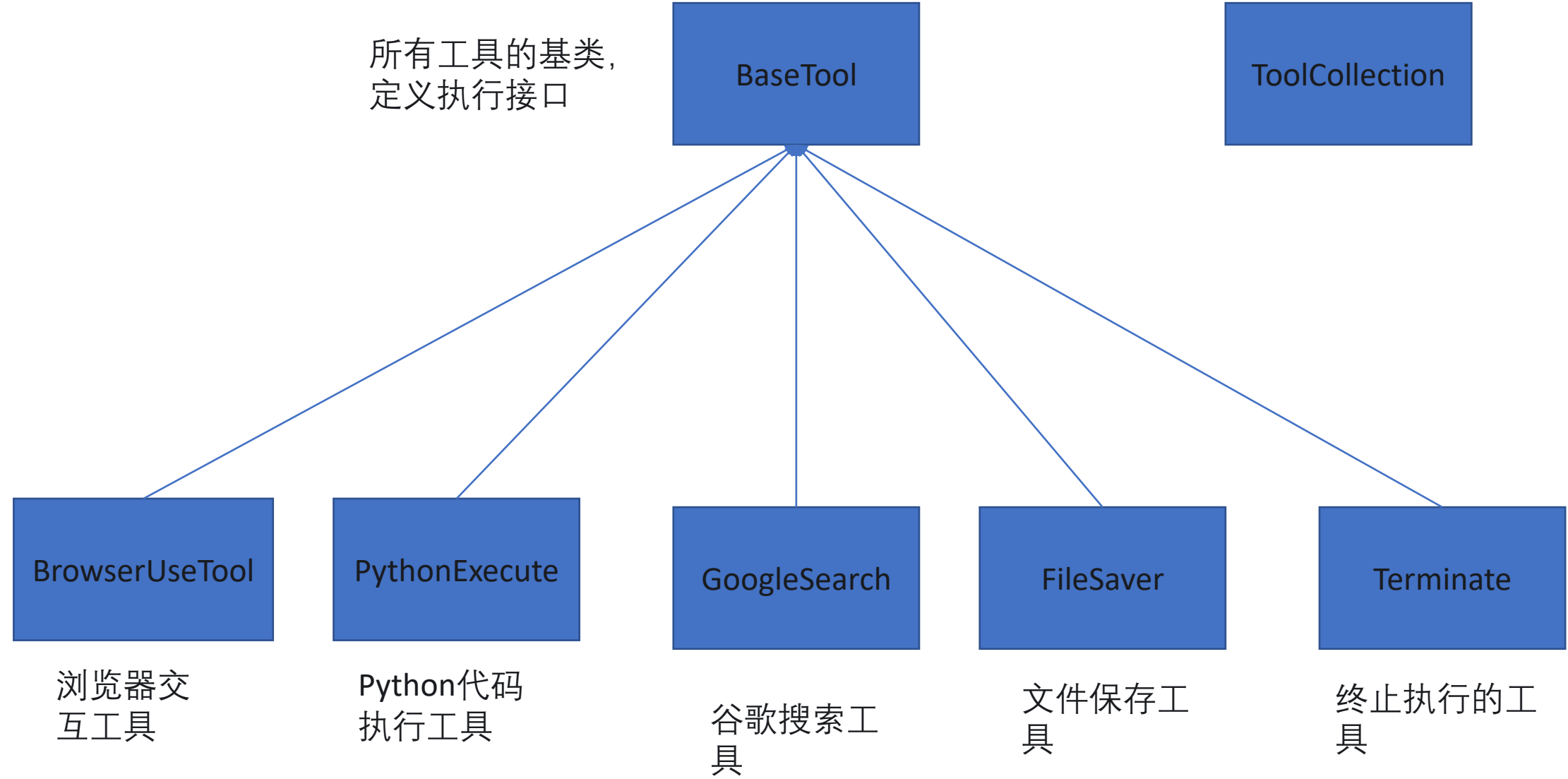
整体架构

```
OpenManus/
├── app/                                # 主要应用代码
│   ├── agent/                          # 代理类实现
│   ├── prompt/                         # 系统提示和指令
│   ├── tool/                           # 工具类实现
│   └── llm.py                          # 语言模型交互
│   ├── logger.py                       # 日志系统
│   └── schema.py                       # 数据结构定义
├── assets/                             # 静态资源
├── config/                             # 配置文件
├── main.py                             # 主入口
├── run_flow.py                         # 开发版入口
├── requirements.txt                    # 依赖列表
└── setup.py                           # 安装脚本
```

智能体关系图



工具类



用户输入

```
prompt = "查找关于人工智能的最新新闻并保存到文件"
```

在main.py中处理

```
await agent.run(prompt)
```

在BaseAgent.run中添加到内存

```
self.update_memory("user", request)
```

问题分解：

- 1.搜索人工智能
- 2.将内容保存至文件

第一次思考

```
# 获取带工具选项的响应
# Get response with tool options
response = await self.llm.ask_tool([
    messages=self.messages,
    system_msgs=[Message.system_message(self.system_prompt)]
    if self.system_prompt
    else None,
    tools=self.available_tools.to_params(),
    tool_choice=self.tool_choices,
])
self.tool_calls = response.tool_calls
```

大模型输入

```
[Message(role='user', content='查找关于人工智能的最新新闻并保存到文件', tool_calls=None, name=None, tool_call_id=None), Message(role='user', content='You can interact with the computer using PythonExecute, save important content and information files through FileSaver, open browsers with BrowserUseTool, and retrieve information using GoogleSearch.\n\nPythonExecute: Execute Python code to interact with the computer system, data processing, automation tasks, etc.\n\nFileSaver: Save files locally, such as txt, py, html, etc.\n\nBrowserUseTool: Open, browse, and use web browsers. If you open a local HTML file, you must provide the absolute path to the file.\n\nGoogleSearch: Perform web information retrieval\n\nBased on user needs, proactively select the most appropriate tool or combination of tools. For complex tasks, you can break down the problem and use different tools step by step to solve it. After using each tool, clearly explain the execution results and suggest the next steps.\n', tool_calls=None, name=None, tool_call_id=None)]
ChatCompletionMessage(content='', refusal=None, role='assistant', audio=None, function_call=None, tool_calls=[ChatCompletionMessageToolCall(id='call_3ddfc76d346c4840842b3f', function=Function(arguments='{"query": "latest news on artificial intelligence", "num_results": 5}', name='google_search'), type='function', index=0)])
```

大模型输出

```
ChatCompletionMessage(content='', refusal=None, role='assistant', audio=None, function_call=None, tool_calls=[ChatCompletionMessageToolCall(id='call_3ddfc76d346c4840842b3f', function=Function(arguments='{"query": "latest news on artificial intelligence", "num_results": 5}', name='google_search'), type='function', index=0)])
```

运行搜索引擎

```
result = await self.execute_tool(command)
logger.info(
    f"🎯 Tool '{command.function.name}' completed its mission! Result: {result}"
)

# Add tool response to memory
tool_msg = Message.tool_message(
    content=result, tool_call_id=command.id, name=command.function.name
)
self.memory.add_message(tool_msg)
```

将结果添加到内存

```
tool_msg = Message.tool_message(
    content="Search results for 'latest artificial intelligence news':\n...",
    tool_call_id="call_123",
    name="google_search"
)
```

第二次思考

```
# 获取带工具选项的响应
# Get response with tool options
response = await self.llm.ask_tool(
    messages=self.messages,
    system_msgs=[Message.system_message(self.system_prompt)]
    if self.system_prompt
    else None,
    tools=self.available_tools.to_params(),
    tool_choice=self.tool_choices,
)
self.tool_calls = response.tool_calls
```

```
[Message(role='user', content='查找关于人工智能的最新新闻并保存到文件', tool_calls=None, name=None, tool_call_id=None), Message(role='user', content='You can interact with the computer using PythonExecute, save important content and information files through FileSaver, open browsers with BrowserUseTool, and retrieve information using GoogleSearch.\n\nPythonExecute: Execute Python code to interact with the computer system, data processing, automation tasks, etc.\n\nFileSaver: Save files locally, such as txt, py, html, etc.\n\nBrowserUseTool: Open, browse, and use web browsers.If you open a local HTML file, you must provide the absolute path to the file.\n\nGoogleSearch: Perform web information retrieval\n\nBased on user needs, proactively select the most appropriate tool or combination of tools. For complex tasks, you can break down the problem and use different tools step by step to solve it. After using each tool, clearly explain the execution results and suggest the next steps.\n', tool_calls=None, name=None, tool_call_id=None), Message(role='assistant', content='', tool_calls=[ToolCall(id='call_3ddfc76d346c4840842b3f', type='function', function=Function(name='google_search', arguments={'query': 'latest news on artificial intelligence', 'num_results': 5}))]), name=None, tool_call_id=None), Message(role='tool', content="Observed output of cmd `google_search` executed:\n['https://www.wsj.com/tech/ai', 'https://www.bbc.com/news/topics/ce1qrvleqt', 'https://www.artificialintelligence-news.com/', 'https://www.nbcnews.com/artificial-intelligence', 'https://finance.yahoo.com/news/3-scorching-hot-artificial-intelligence-085100134.html']", tool_calls=None, name='google_search', tool_call_id='call_3ddfc76d346c4840842b3f'), Message(role='user', content='You can interact with the computer using PythonExecute, save important content and information files through FileSaver, open browsers with BrowserUseTool, and retrieve information using GoogleSearch.\n\nPythonExecute: Execute Python code to interact with the computer system, data processing, automation tasks, etc.\n\nFileSaver: Save files locally, such as txt, py, html, etc.\n\nBrowserUseTool: Open, browse, and use web browsers.If you open a local HTML file, you must provide the absolute path to the file.\n\nGoogleSearch: Perform web information retrieval\n\nBased on user needs, proactively select the most appropriate tool or combination of tools. For complex tasks, you can break down the problem and use different tools step by step to solve it. After using each tool, clearly explain the execution results and suggest the next steps.\n', tool_calls=None, name=None, tool_call_id=None)]
```

```
ChatCompletionMessage(content='我找到了一些关于人工智能的最新新闻链接：\n\n1. [Aibase](https://www.aibase.com/zh/news)\n2. [OFweek人工智能网](https://www.ofweek.com/ai/CATList-201700-8100-ai.html)\n3. [BBC中文网](https://www.bbc.com/zhongwen/articles/c99n7x5jrygo/trad)\n4. [香港01](https://www.hk01.com/tag/3224)\n5. [IT之家](https://www.ithome.com/tags/%E4%BA%BA%E5%B7%A5%E6%99%BA%E8%83BD/)\n\n接下来，我将保存这些链接到一个文本文件中。', refusal=None, role='assistant', audio=None, function_call=None, tool_calls=[ChatCompletionMessageToolCall(id='call_975a2c0ae73c48cca224f5', function=Function(arguments={'content': 'https://www.aibase.com/zh/news\\nhttps://www.ofweek.com/ai/CATList-201700-8100-ai.html\\nhttps://www.bbc.com/zhongwen/articles/c99n7x5jrygo/trad\\nhttps://www.hk01.com/tag/3224\\nhttps://www.ithome.com/tags/%E4%BA%BA%E5%B7%A5%E6%99%BA%E8%83BD/', "file_path": "ai_news_links.txt", "mode": "w"}), name='file_save'), type='function', index=0)])
```

执行FileSaver工具

```
result = await self.execute_tool(command)
```

FileSaver.execute方法

```
async def execute(self, content: str, file_path: str, mode: str = "w") -> ToolResult:
```

```
    with open(file_path, mode, encoding="utf-8") as f:
```

```
        f.write(content)
```

```
    return ToolResult(output=f"Content successfully saved to {file_path}")
```

将结果添加到内存

```
tool_msg = Message.tool_message(
```

```
    content="Content successfully saved to ai_news.txt",
```

```
    tool_call_id="call_456",
```

```
    name="file_saver"
```

```
)
```

```
self.memory.add_message(tool_msg)
```


第三次思考

任务已经完成， 结束

```
[Message(role='user', content='查找关于人工智能的最新新闻并保存到文件', tool_calls=None, name=None, tool_call_id=None), Message(role='user', content='You can interact with the computer using PythonExecute, save important content and information files through FileSaver, open browsers with BrowserUseTool, and retrieve information using GoogleSearch.\n\nPythonExecute: Execute Python code to interact with the computer system, data processing, automation tasks, etc.\n\nFileSaver: Save files locally, such as txt, py, html, etc.\n\nBrowserUseTool: Open, browse, and use web browsers.If you open a local HTML file, you must provide the absolute path to the file.\n\nGoogleSearch: Perform web information retrieval\n\nBased on user needs, proactively select the most appropriate tool or combination of tools. For complex tasks, you can break down the problem and use different tools step by step to solve it. After using each tool, clearly explain the execution results and suggest the next steps.\n', tool_calls=None, name=None, tool_call_id=None), Message(role='assistant', content='', tool_calls=[ToolCall(id='call_9dee64d3533e4345a8e303', type='function', function=Function(name='google_search', arguments='{"query": "最新的人工智能新闻", "num_results": 5}')]), name=None, tool_call_id=None), Message(role='tool', content='Observed output of cmd `google_search` executed:\n[\'https://www.aibase.com/zh/news\', \'https://www.ofweek.com/ai/CATList-201700-8100-ai.html\', \'https://www.bbc.com/zhongwen/articles/c99n7x5jrygo/trad\', \'https://www.hk01.com/tag/3224\', \'https://www.ithome.com/tags/%E4%BA%BA%E5%B7%A5%E6%99%BA%E8%83%BD/\']', tool_calls=None, name='google_search', tool_call_id='call_9dee64d3533e4345a8e303'), Message(role='user', content='You can interact with the computer using PythonExecute, save important content and information files through FileSaver, open browsers with BrowserUseTool, and retrieve information using GoogleSearch.\n\nPythonExecute: Execute Python code to interact with the computer system, data processing, automation tasks, etc.\n\nFileSaver: Save files locally, such as txt, py, html, etc.\n\nBrowserUseTool: Open, browse, and use web browsers.If you open a local HTML file, you must provide the absolute path to the file.\n\nGoogleSearch: Perform web information retrieval\n\nBased on user needs, proactively select the most appropriate tool or combination of tools. For complex tasks, you can break down the problem and use different tools step by step to solve it. After using each tool, clearly explain the execution results and suggest the next steps.\n', tool_calls=None, name=None, tool_call_id=None), Message(role='assistant', content='我找到了一些关于人工智能的最新新闻链接：\n\n1. [AIbase](https://www.aibase.com/zh/news)\n2. [OFweek人工智能网](https://www.ofweek.com/ai/CATList-201700-8100-ai.html)\n3. [BBC中文网](https://www.bbc.com/zhongwen/articles/c99n7x5jrygo/trad)\n4. [香港01](https://www.hk01.com/tag/3224)\n5. [IT之家](https://www.ithome.com/tags/%E4%BA%BA%E5%B7%A5%E6%99%BA%E8%83%BD/)\n\n接下来，我将保存这些链接到一个文本文件中。', tool_calls=[ToolCall(id='call_975a2c0ae738cca224f5', type='function', function=Function(name='file_saver', arguments='{"content": "https://www.aibase.com/zh/news\\nhttps://www.ofweek.com/ai/CATList-201700-8100-ai.html\\nhttps://www.bbc.com/zhongwen/articles/c99n7x5jrygo/trad\\nhttps://www.hk01.com/tag/3224\\nhttps://www.ithome.com/tags/%E4%BA%BA%E5%B7%A5%E6%99%BA%E8%83%BD/", "file_path": "ai_news_links.txt", "mode": "w"}')]), name=None, tool_call_id=None), Message(role='tool', content='Observed output of cmd `file_saver` executed:\nContent successfully saved to ai_news_links.txt', tool_calls=None, name='file_saver', tool_call_id='call_975a2c0ae738cca224f5'), Message(role='user', content='You can interact with the computer using PythonExecute, save important content and information files through FileSaver, open browsers with BrowserUseTool, and retrieve information using GoogleSearch.\n\nPythonExecute: Execute Python code to interact with the computer system, data processing, automation tasks, etc.\n\nFileSaver: Save files locally, such as txt, py, html, etc.\n\nBrowserUseTool: Open, browse, and use web browsers.If you open a local HTML file, you must provide the absolute path to the file.\n\nGoogleSearch: Perform web information retrieval\n\nBased on user needs, proactively select the most appropriate tool or combination of tools. For complex tasks, you can break down the problem and use different tools step by step to solve it. After using each tool, clearly explain the execution results and suggest the next steps.\n', tool_calls=None, name=None, tool_call_id=None), Message(role='assistant', content='这些链接已经成功保存到名为 `ai_news_links.txt` 的文本文件中。如果您需要进一步的信息或执行其他任务，请告诉我。', tool_calls=None, name=None, tool_call_id=None), Message(role='user', content='You can interact with the computer using PythonExecute, save important content and information files through FileSaver, open browsers with BrowserUseTool, and retrieve information using GoogleSearch.\n\nPythonExecute: Execute Python code to interact with the computer system, data processing, automation tasks, etc.\n\nFileSaver: Save files locally, such as txt, py, html, etc.\n\nBrowserUseTool: Open, browse, and use web browsers.If you open a local HTML file, you must provide the absolute path to the file.\n\nGoogleSearch: Perform web information retrieval\n\nBased on user needs, proactively select the most appropriate tool or combination of tools. For complex tasks, you can break down the problem and use different tools step by step to solve it. After using each tool, clearly explain the execution results and suggest the next steps.\n', tool_calls=None, name=None, tool_call_id=None), Message(role='assistant', content='', tool_calls=[ToolCall(id='call_00ab67a44a054b53ab56d1', type='function', function=Function(name='terminate', arguments='{"status": "success"}')]), name='terminate', tool_call_id='call_00ab67a44a054b53ab56d1')]
```

```
ChatCompletionMessage(content='', refusal=None, role='assistant', audio=None, function_call=None, tool_calls=[ChatCompletionMessageToolCall(id='call_00ab67a44a054b53ab56d1', function=Function(arguments='{"status": "success"}', name='terminate'), type='function', index=0)])
```